

Research paper

A virtual 3D printing framework for off-Earth construction

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ABSTRACT

3D printing, due to its digital nature, offers a practical approach to autonomous off-Earth construction by utilizing local materials, thereby reducing the reliance on Earth-based resources and manpower. Implementing large-scale and rapid 3D printing of structures in extraterrestrial environments poses unique challenges due to conditions such as altered gravity, vacuum, and extreme thermal variations. These challenges are compounded by the fact that replicating altered gravity on Earth is not feasible for extended periods of time. Therefore, a virtual system is needed to advance the 3D printing capabilities. This paper presents computational fluid dynamics (CFD) model, validated through experimental 3D printing of lunar regolith simulant-based geopolymer on Earth. The model is used to simulate the 3D printing of structures under altered gravity conditions. The results provide insights into geometrical precision and deformation as gravity varies across different celestial bodies, as well as the influence of regolith mortar properties and nozzle diameter when focusing on printing on the Moon. The results highlight ways to control the curing kinetics of the lunar regolith mortar in order to print fast and on large scale.

Understanding construction under different gravity levels represents a transformative step towards realizing human dreams of colonizing and exploring other planets. The first step is establishing a self-sustaining lunar outpost, which has long been a key objective in advancing human space exploration [1,2]. Building infrastructure like bases, landing pads, and habitats requires various protective measures, including radiation protection, thermal insulation, exhaust plume containment, and micro-meteoroid shielding. Therefore, innovative solutions are essential from both material science and engineering perspectives. 3D printing technology was conceptualized for the construction of lunar habitats back in 2005 [3,4]. On Earth, 3D printing has demonstrated considerable potential in construction due to its unparalleled versatility and efficiency in fabricating complex structures. Given its digital nature, this technology is highly promising for off-Earth construction, where minimal supervision will be required. There are, however, a variety of challenges arising from both the material and printing processes, as well as unknown structural behavior under altered environmental conditions.

Transporting materials from Earth is impractical, underscoring the imperative of in-situ resource utilization (ISRU) [5,6]. The regolith available on the celestial bodies is the obvious choice for off-Earth construction. For instance, lunar regolith serves as a raw material for producing aggregates [7], binders [8,9], and reinforcements [3,10], ensuring its versatility. Notably, National Aeronautics and Space Administration (NASA) supported projects focuses on the use of 3D printing

with lunar regolith simulants to create the Lunar Plume Alleviation Device [11,12]. Similarly, the European Space Agency (ESA) proposed a visionary concept in 2016 entitled “Moon Village” [13], advocating for a permanent lunar surface outpost developed through international collaboration. For extrusion-based 3D printing, regolith can be processed into feedstock composition in the form of filament, paste, or mortar using methods such as polymer-based regolith composites [14], sulfur concrete [6], and geopolymers [9]. These methods involve several stages, including crushing and grinding, binder selection and mixing, and rheology modification with additives and thixotropic agents [15, 16] before the actual printing process. Rheological characterization of the feedstock is essential for ensuring extrudability and identifying issues, such as interlayer bonding, that could affect print quality.

Environmental conditions in space, including altered gravity, extreme temperature gradients, unique heat flow dynamics, and vacuum conditions, can significantly impact the 3D printing process. A limited number of studies were able to consider 3D printing under some of these conditions. To investigate the influence of microgravity during filament-based extrusion 3D printing, experimental efforts have taken place on board the ISS [17] and in parabolic flights [18,19]. These tests revealed issues like defects in printed layers and challenges with layer adhesion in microgravity [18,19]. On Earth, vacuum tests for filament-based printing showed defects such as bubbling and layer damage [20,21]. For structural applications, studies on mortar/paste

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in vacuum highlight significant water loss through evaporation [22], complicating continuous regolith extrusion [23]. Despite these crucial insights, experimental setups are complex and rare, especially in altered gravity, emphasizing the need for numerical modeling to virtually understand 3D printing under off-Earth conditions, particularly in the presence of altered gravity.

The efficacy of the layer-by-layer deposition technique heavily relies on implementing suitable printing strategies, which directly influence the morphological characteristics of the layers as understood from the printing on the Earth [24–26]. During deposition, it is imperative that the layers maintain geometric stability to avoid defects such as excessive deformation, weak interlayer bonding, and poor surface roughness, thereby ensuring the structural integrity and overall quality of the deposited part [27–29].

Various printing strategies have been investigated in the scientific literature, focusing on printing on the Earth. (i) Wet-on-wet printing involves depositing layers onto wet layers without giving enough time for the material to change and start curing. Studies have shown that in wet-on-wet printing, although adjustments to material and processing parameters can reduce layer deformation to some extent, however, the deformation cannot be eliminated entirely [30,31]. (ii) Wet-on-solid printing refers to depositing layers onto fully solidified layers. Research showed that wet-on-solid printing illustrates stable printing; however, it can cause weak interlayer bonding and prolonged printing duration due to solidification [27,32]. (iii) Wet-on-semisolid printing allows previously deposited layers to solidify to a certain degree before printing the subsequent layer [33]. This strategy has been shown to control deformation with a yield stress buildup, requiring a moderately long time. Faster printing can be achieved by effectively utilizing the continuous buildup of material, which enables the use of low-yield-stress materials and is expected to enhance inter-layer bonding while maintaining stability through controlled deformation.

The deformation described refers to the distortion of a previously deposited layer caused by the deposition of a new layer on top of it. This occurs due to the total applied pressure, which includes both extrusion pressure and hydrostatic pressure (the weight of the deposited material) during printing. After printing, the hydrostatic pressure remains active and can cause continuous deformation if the material's yield stress is not sufficiently large. Given the persistent risk of deformation and its relation to the applied pressure, the printing process must consider the specific gravity magnitudes to investigate the geometrical precision and control of the deformation of the deposited part.

This study looks into 3D printing under altered gravity using Computational Fluid Dynamics (CFD) modeling. Furthermore, we investigate the mortar properties required to achieve stable prints and geometrically precise layers for 3D printing on the Moon. In addition to typical wet-on-wet printing, we explored wet-on-solidifying printing, which accounts for the curing of the material during the printing process. The curing is modeled as a material age-dependent yield stress increase. This approach can optimize printing speed while maintaining stability and interlayer bonding. It can also open the possibility of designing materials with specific curing kinetics to support both stable and rapid printing.

1. Methodology for numerical framework

1.1. Materials and rheology

When it comes to 3D printing for the construction of lunar habitats, the rheological characteristics of lunar regolith mortar are crucial. Compared with traditional building materials, lunar regolith has unique flow properties due to its fine-grained dust and rocky fragments. Factors such as particle size distribution, shape, and surface roughness affect the rheological behavior of the mortar obtained from the lunar regolith [34], similar to typical mortar/paste for on-Earth constructions [35–37].

In the process of creating a printable paste, lunar regolith is required to blend with a binding agent. In this process, lunar simulant regolith is typically used that replicates the chemical and physical properties of lunar regolith. This study focuses on the lunar regolith simulant JSC-1 A, which was produced through the milling and sieving of basalt cinders originating from Merriam Crater in Arizona [38]. The simulant JSC-1 A includes a compatible amount of elemental oxide as found in Apollo samples [39]. Momi et al. [34] examined the rheological properties of samples comprising simulant regolith JSC-1 A, water, NaOH (an alkaline agent), and urea (a superplasticizer). It was found that the rheological properties of the samples exhibit viscoplastic behavior and can be predicted well with the Bingham rheological model [40]. The Bingham model assumes that until the applied shear stress τ exceeds the yield stress τ_0 , the material remains solid and does not flow. Once the applied stress surpasses τ_0 , the material starts flowing linearly with the shear rate. The viscosity μ of the material is a function of shear rate $\dot{\gamma}$, and is expressed as:

$$\mu(\dot{\gamma}) = \begin{cases} \infty, & \text{when } \tau < \tau_0 \\ \eta_p + \frac{\tau_0}{\dot{\gamma}}, & \text{when } \tau \geq \tau_0 \end{cases} \quad (1)$$

where η_p is the plastic viscosity of the material.

When the material solidifies over time during the printing process, the material builds its yield stress. In that case, the building yield stress is $\tau_B(T_c) = \alpha T_c + \tau_0$; where α is the curing factor and T_c is the curing time of the material. Therefore, the viscosity of the material is a function of both shear rate and time, and can be presented by the Eq. 1.1 when replacing τ_0 by τ_B . The viscosity becomes infinite when $\tau < \tau_B$ or at zero shear rates, which typically causes numerical instabilities in the computational model. Therefore, the following bi-viscous regularization [41] of the Bingham material model is used, where the apparent viscosity is as follows:

$$\mu_{app}(\dot{\gamma}, T_c) = \begin{cases} \mu_{M,app}, & \text{when } \dot{\gamma} < \dot{\gamma}_c \\ \eta_p + \frac{\tau_B}{\dot{\gamma}}, & \text{when } \dot{\gamma} \geq \dot{\gamma}_c \end{cases} \quad (2)$$

where $\mu_{M,app} = \eta_p + \frac{\tau_B}{\dot{\gamma}_c}$ is the maximum apparent viscosity and $\dot{\gamma}_c$ is the critical shear rate. The choice of $\dot{\gamma}_c$ is arbitrary, yet it should be sufficiently small to lead to a relatively higher maximum apparent viscosity to restrict flow in regions where stress is below the yield stress, yet sizable enough to prevent prolonged computations and potential divergence problems. In this study, $\dot{\gamma}_c = 0.10$ and 0.03 s^{-1} in accordance with the Refs. [30,33]. It should be noted that the bi-viscous regularization of the Bingham constitutive model may influence the solution, as it employs a finite maximum apparent viscosity based on the critical shear rate.

The details of the samples considered for this study and their rheological properties are summarized in Table 1.

1.2. Governing equations

The extrusion-deposition flow of the regolith simulant is assumed to be transient, isothermal, and incompressible with a constant density ρ . Therefore, the flow dynamics are governed by the following mass and momentum conservation equations:

$$\nabla \cdot \mathbf{u} = 0 \quad (3)$$

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \rho \mathbf{g} + \nabla \cdot \boldsymbol{\sigma} \quad (4)$$

where $\mathbf{u}$ is the velocity vector, p is the pressure, t is the time, $\boldsymbol{\sigma}$ is the deviatoric stress tensor, and $\mathbf{g}$ is the gravitational acceleration vector with components $(0, 0, -g)$. Here, g varies based on the gravity of the investigated celestial bodies; for instance, $g = 1.62 \text{ m/s}^2$ is the gravity of the Moon.

The deviatoric stress tensor is approximated with the following generalized Newtonian constitutive model using the apparent viscosity function described in the previous section:

$$\boldsymbol{\sigma} = 2\mu_{app}(\dot{\gamma}, T_c) \mathbf{D}_T \quad (5)$$

Table 1

The details of the samples considered for this study and their rheological properties, as obtained from Momi et al. [34].

Cases	JSC-1A regolith (g)	Water (g)	Regolith content w_R (wt%)	Volume fraction (%)	Density ρ (kg/m ³)	Dynamic yield stress τ_0 (Pa)	Plastic viscosity η_p (Pa.s)
1	60	20.4	74.6	50.0	1970	9.5	0.6
2	60	19.4	75.8	51.3	1995	18.0	1.3
3	60	18.6	76.3	52.3	2015	26.0	2.4
4	60	18.0	76.9	53.1	2031	36.0	3.4
5	60	17.6	77.3	53.7	2042	41.0	4.4

where $\dot{\gamma} = \sqrt{2\text{tr}(\mathbf{D}_T^2)}$ represents the strain rate magnitude and $\mathbf{D}_T$ represents the deformation rate tensor, defined as:

$$\mathbf{D}_T = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) \quad (6)$$

where the superscript T stands for the transpose notation.

1.3. CFD model in a nutshell

Two CFD models have been developed to simulate the deposition of regolith in microgravity: Model 1 simulates the wet-on-wet printing strategy (when the curing factor is zero), and Model 2 simulates wet-on-solidifying. Both models comprise a cylindrical extrusion nozzle, a substrate/build plate, and a top surface, as seen in Fig. 1. The extrusion nozzle of diameter D mm is placed at a distance h mm from the substrate (also known as the gap between the nozzle and substrate). The nozzle deposits material and forms a layer of nominal height h . During the deposition, the material is supplied at an extrusion speed U mm/s, and the nozzle moves with a printing speed V mm/s. The details of printing conditions and the computational domain are summarized in Table 2.

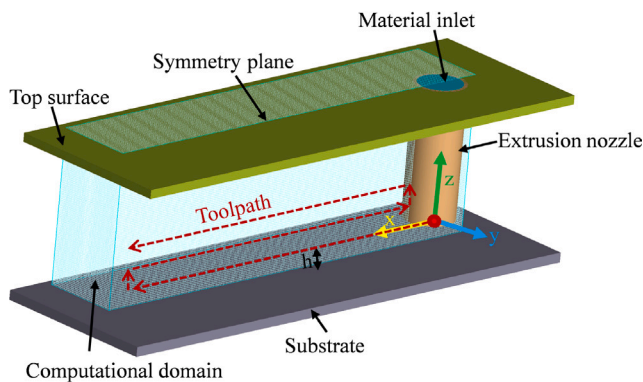


Fig. 1. CFD model geometry with toolpath.

A symmetrical boundary condition is applied along the xz -plane of the computational domain to reduce computational cost. The top plane acts as the material inlet, where the top surface (an artificial solid object) is placed to prevent material flow from cells outside the nozzle orifice. The bottom plane is applied to a wall boundary condition that holds the solid substrate. Other planes are set to continuous boundary conditions. Furthermore, the solid-fluid interaction is subjected to a no-slip boundary condition.

A uniform Cartesian grid is used to mesh the computational domain. The mesh sensitivity was evaluated for the reference model using cell sizes of 0.9, 1.0, and 1.1 mm, corresponding to total cell counts of 1.36, 1.00, and 0.75 million, respectively. The cross-sectional shapes of the deposited layers for the different mesh sizes are compared in Fig. 8 (Appendix), and the quantitative measures, such as the total height of the deposited part and the width of the bottom layer, are summarized in Table 5 in Appendix. Only negligible differences were observed across the tested meshes; therefore, considering the balance between accuracy and computational efficiency, a cell size of 1.0 mm was selected for the simulations. The governing equations (1) and (2) are discretized over the computational domain using the Finite Volume Method and solved using the CFD tool FLOW-3D[®] (V22R1; Flow Science Inc.) [42]. To embed solid objects (nozzle, substrate, etc.) in the computational domain, FLOW-3D[®] uses the FAVOR (Fractional Area/Volume Obstacle Representation) method. It facilitates material extrusion through the nozzle that moves and deposition onto the substrate by incorporating solid-geometry effects into the governing equations so that solid objects can be predicted accurately. Furthermore, the used Immersed Boundary Methods facilitate the prediction of pressure and forces near solid objects [43]. An explicit upwind-difference technique ensures second-order accuracy in space and first-order accuracy in time in calculating momentum advection. An implicit solver called GMRES (Generalized Minimum Residual) is utilized to solve the pressure-velocity components of the governing equations [44–46]. The viscous stress of the momentum equation is solved by an implicit method. The free surface tracking is carried out using the Volume-of-Fluid technique [47,48]. During simulations, the time step size is controlled dynamically based on the obtained stability limit to avoid numerical instabilities, however, an initial minimum value is set at 1×10^{-5} [42]. For further information, one can be referred to the modeling sections of Refs. [30,33,49].

Table 2

The details of the CFD models. Note that the size of the computational domain varies with D , and model 1 and model 2 utilize critical shear rates $\dot{\gamma}_c = 0.10$ and 0.03 s^{-1} , respectively.

Model ID	Nozzle diameter D (mm)	No. of deposited layers N (–)	Length of deposited layer (mm)	Printing speed (mm/s)	Extrusion speed (mm/s)	Volumetric flow rate (mm ³ /s)	Printing time (s)	Curing time (s)	Curing factor (–)	Size of computational domain (mm ³)
1	5	3	$8D$	40	40	$7.854\text{e}+2$	3.188	–	–	$9D \times 2D \times (3D + h)$
1	15	3	$8D$	40	40	$7.07\text{e}+3$	9.563	–	–	$9D \times 2D \times (3D + h)$
1	25	3	$8D$	40	40	$1.964\text{e}+4$	15.938	–	–	$9D \times 2D \times (3D + h)$
1	50	3	$8D$	40	40	$7.854\text{e}+4$	31.875	–	–	$9D \times 2D \times (3D + h)$
2	50	2	$5D$	40	40	$7.854\text{e}+4$	135.25	135.25	0	$6D \times 2D \times (2D + h)$
2	50	2	$5D$	40	40	$7.854\text{e}+4$	135.25	135.25	0.92	$6D \times 2D \times (2D + h)$
2	50	2	$5D$	40	40	$7.854\text{e}+4$	135.25	135.25	4.59	$6D \times 2D \times (2D + h)$

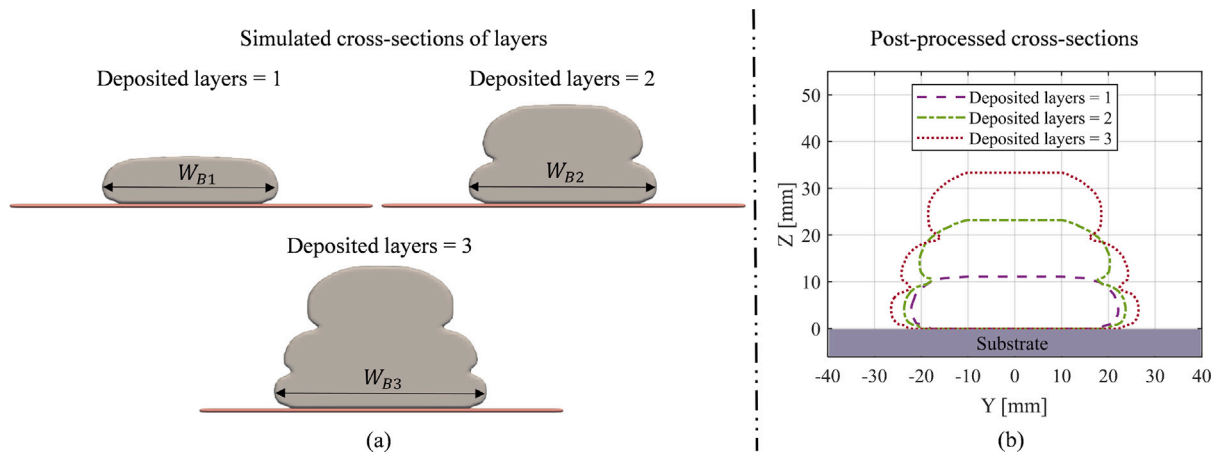


Fig. 2. Post-processing of simulated results: (a) simulated cross-sections and the definition of the width of the bottom layer, and (b) post-processed cross-sections.

1.4. Results post-processing

The geometrical stability of the prints and the extrusion pressure are analyzed by importing the fluid fraction and pressure field from the CFD software into MATLAB. The cross-sections of layers are extracted at the middle of the deposited layer, as shown in Fig. 2. The relative percentile deformation of the first layer after printing both layers is calculated by:

$$D_B = \frac{W_{Bi} - W_{B1}}{W_{B1}} \times 100\% \quad (7)$$

where W_{B1} is the width of the bottom layer after its deposition and W_{Bi} is the width of the i th layer after its deposition, where $i = 2$ and 3. Thus, D_B is required to be small for stable printing.

2. Experimental methodology and CFD model validation

2.1. Rheology measurements and 3D printing

An experimental campaign was conducted to validate the CFD model. Three successive layers were printed using a geopolimer formulated with lunar mare regolith simulant (LMS-1) as the primary binder and activated by an alkaline solution consisting of 8 M sodium hydroxide and sodium silicate in a mass ratio of 2:15, as described in previous studies with geopolymers [50,51]. A binder-to-activator mass ratio of 0.41 was maintained throughout the experiments. The oxide composition of the mare regolith (in wt%) is presented in Table 3. Prior to rheology measurements and 3D printing, LMS-1 was milled to a D50 particle size of $\approx 10 \mu\text{m}$.

Table 3

Oxide composition of LMS-1 (wt%) [52].

SiO ₂	TiO ₂	Al ₂ O ₃	FeO	MnO	MgO	CaO	Na ₂ O	K ₂ O	P ₂ O ₅
47.42	2.68	14.02	8.74	0.18	14.91	8.26	1.72	0.39	0.20

The rheology of the lunar regolith geopolimer was characterized using an Anton Paar MCR 502 rotational rheometer. A parallel plate configuration was used for the measurements, with a plate diameter of 25 mm and a plate gap of 2 mm. Several controlled shear rate tests were conducted to obtain the flow curve of the mixture. Each test began with a 50-second pre-shearing step consisting of five successive strain levels ranging from 10% to 50%, applied at 1 Hz for 10 s each. This was followed by a stepwise shear rate ramp from 0.1 s^{-1} to 8 s^{-1} , logarithmically from 0.1 to 1 s^{-1} , and linearly from 1 to 8 s^{-1} . The ramp-down was then performed in reverse, all under isothermal conditions at 25°C . All samples were allowed a setting time of approximately 20 min from the moment the activator was added to the binder, consistent with the setting time prior to printing.

A clear hysteresis loop between the ramp-up and ramp-down shear curves was observed, indicating time-dependent structural changes during shear, similar to what is found in the literature for the Martian regolith, cf. [53]. The ramp-down data were fitted using linear regression based on least-squares minimization, and obtained the yield stress and plastic viscosity for the nine test cases of the same composition. It is observed that the yield stress and plastic viscosity of the mixtures vary across the test cases. This variation may be attributed to factors such as the heterogeneous particle size distribution of the lunar regolith and minor procedural inconsistencies, including differences in mixing and setting times. To provide representative behavior, a global fit was performed using data from all nine test cases. This resulted in a yield stress of $329.51 \pm 44.4 \text{ Pa}$ and a plastic viscosity of $73.46 \pm 8.3 \text{ Pa}\cdot\text{s}$.

The experimental study was conducted using a HYREL 3D printer with an SR motor, equipped with a piston cartridge of 60 ml capacity and operated via Repetrel software. The material was extruded through a nozzle with an orifice diameter of $\phi 4.75 \text{ mm}$, positioned 4.5 mm above the build plate. The nozzle speed (i.e., the printing speed) was set to 3.33 mm/s . The extrusion speed was 5.15 mm/s , as calculated from the experimentally observed material displacement within the syringe. Three successive layers were deposited, as shown in Fig. 3a-left. After printing, the specimen was cured and sliced to obtain the cross-sectional shapes of the single-layer, two-layer, and three-layer prints, as shown in Fig. 3a-right.

2.2. Comparison between experiment and simulations

The CFD model simulated the three layers using the same parametric details as in the experiments, as well as the yield stress and plastic viscosity obtained from the rheological characterization. A comparison between the experimentally obtained and simulated cross-sectional shapes shows that the simulation well predicted the geometry of the deposited layers, cf. Fig. 3b. In particular, the morphology of the single- and two-layer prints was accurately predicted. A slight discrepancy in strand height for the three-layer print was observed, which can be attributed to the solidification of the material during the printing process. On the other hand, the variation in the bottom layer's width resulting from the addition of subsequent layers was well captured.

Furthermore, a quantitative comparison of the cross-sectional area, bottom-layer width, and total height of the printed layers is provided in Table 4. Note that, the width of the bottom layer was measured at the mid-height of the bottom layer for all single-layer, two-layer, and three-layer prints. The percentage errors between the CFD predictions and the averaged experimental measurements were generally low, with cross-sectional area errors below 6%, bottom-layer width errors ranging from approximately 4% to 8%, and height errors remaining within 5%. The RMSE values further confirm the consistency between the simulated

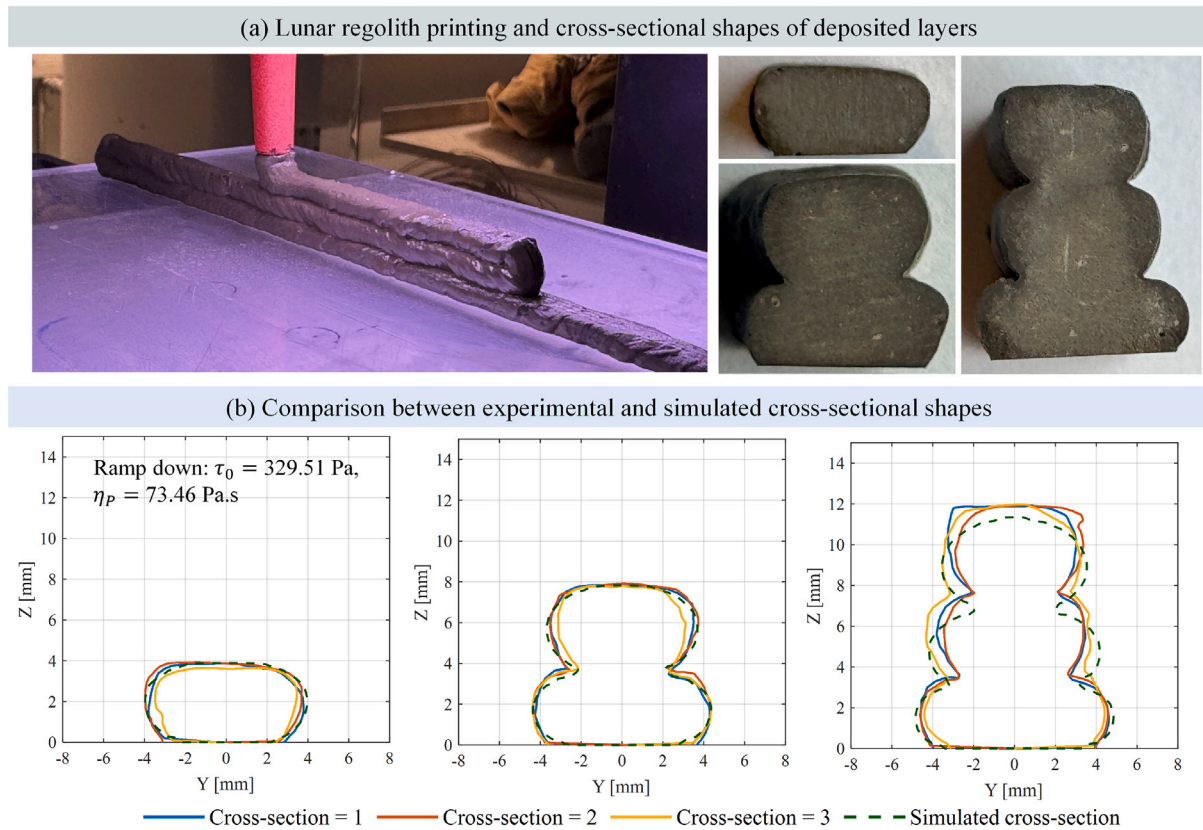


Fig. 3. Validation of the CFD model: (a) 3D printing of Lunar regolith and showing the cross-sectional shapes obtained for up to three layers, and (b) qualitative comparison between the experimental and simulated cross-sectional shapes.

Table 4

Quantitative comparison between experimental measurements and CFD predictions. Here, E_1 , E_2 , and E_3 represent the three experimental cross-sections.

No. of layers	Quantity	E_1 (Exp. C.S. 1)	E_2 (Exp. C.S. 2)	E_3 (Exp. C.S. 3)	S (CFD)	$\bar{E} = \frac{E_1 + E_2 + E_3}{3}$	RMSE = $\sqrt{\frac{1}{3} \sum_{i=1}^3 (S - E_i)^2}$	% Error = $\frac{ S - \bar{E} }{\bar{E}} \times 100\%$
1	Area (mm ²)	26.039	26.891	22.661	26.499	25.197	2.243	5.167
	Width (mm)	7.525	7.642	6.816	7.940	7.328	0.713	8.356
	Height (mm)	3.601	3.880	3.880	3.889	3.787	0.166	2.693
2	Area (mm ²)	54.704	55.854	53.414	53.690	54.657	1.389	1.770
	Width (mm)	8.295	8.418	8.231	8.683	8.315	0.376	4.430
	Height (mm)	7.871	7.920	7.920	8.076	7.904	0.174	2.180
3	Area (mm ²)	80.762	78.918	84.405	80.630	81.362	2.394	0.899
	Width (mm)	9.187	9.180	8.817	9.680	9.061	0.642	6.828
	Height (mm)	11.870	11.920	11.980	11.403	11.923	0.522	4.364

and experimental data across the three repeated cross sections for each layer. Overall, the model demonstrates good agreement with the experimental data, validating its applicability to regolith printing.

3. Results and discussion

3.1. The role of gravity in off-Earth 3D printing

In the context of off-Earth printing, the printed part will experience different applied pressures depending on the variations in gravitational acceleration across different celestial bodies. Consequently, 3D printing under altered gravity may require varied approaches to achieve geometrical precision and control over deformation based on the morphological characteristics of the deposited layers.

To investigate the impact of altered gravity during printing, the CFD Model 1 is used to simulate multilayer printing (three layers in this case) under the gravitational conditions of five different celestial

bodies. The Moon (1.62 m/s²) and Mars (3.71 m/s²) are frequently studied for future human colonization and in-situ construction due to their relative proximity and feasible surface conditions [54–58]. Phobos, a moon of Mars with an extremely low gravitational acceleration of 0.0057 m/s², represents an example of low-gravity environments encountered on small near-Earth asteroids, which could serve as natural way-stations for Earth-to-Mars transportation [59–61]. Furthermore, Ross 128b, an exoplanet with a gravity of approximately 10.95 m/s², is a good candidate because of its Earth-like features and potential habitability [62]. By selecting Phobos, the Moon, Mars, Earth, and Ross 128b, this study encompasses a wide range of gravitational accelerations, both below and above that of Earth, enabling a detailed analysis of how gravity influences layer deposition.

The model utilizes lunar regolith mortar to isolate the effect of gravity. The mortar, as prepared by Momi et al. [34], is a mixture of the regolith simulant JSC-1 A with water into a slurry, yielding a material with a yield stress of 41 Pa and a plastic viscosity of 4.4 Pa.s. The mortar is extruded wet-on-wet using a cylindrical nozzle of

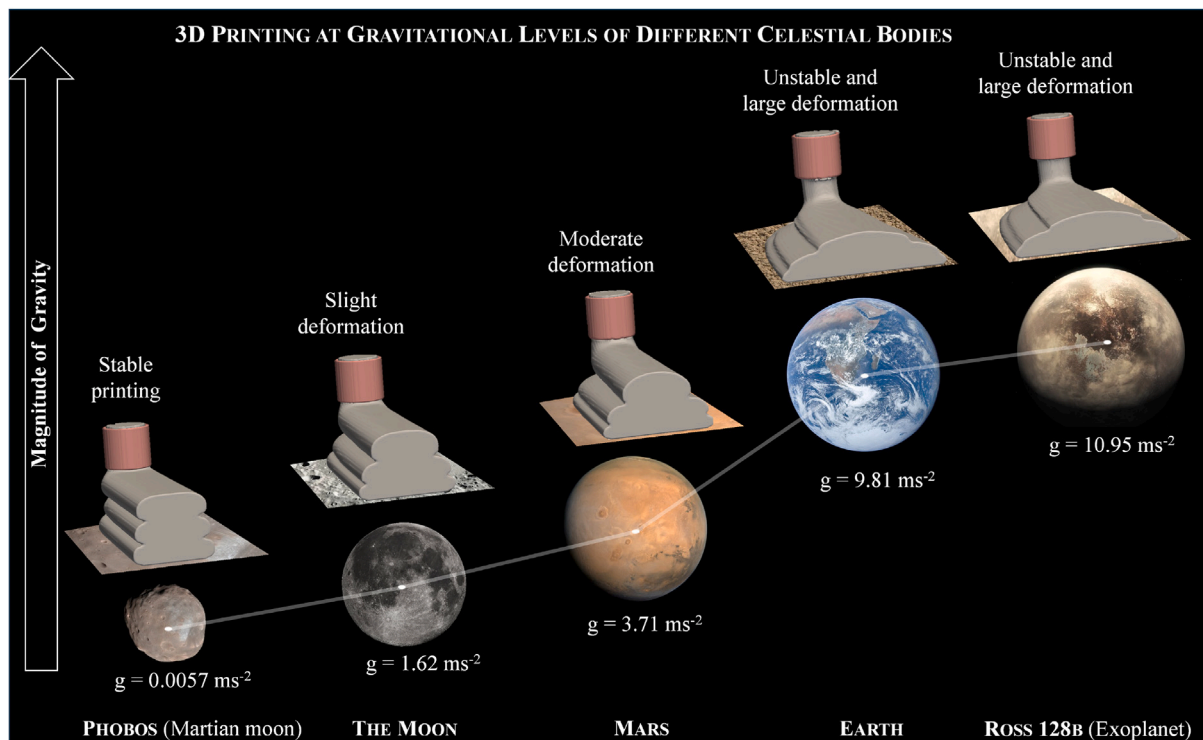


Fig. 4. 3D printing at gravitational levels of different celestial bodies. The same mortar with the same rheological properties is virtually printed with the gravitational acceleration of five different bodies: Phobos (Moon of Mars), the Moon, Mars, Earth, and Ross 128b (Earth-sized exoplanet of the star Ross 128).

$\phi 25$ mm (i.e., nozzle diameter, D) with a nominal layer height $h = D/2$, commonly used for construction purposes on Earth.

Fig. 4 illustrates the printing scenario under varying gravity magnitudes. In higher gravity conditions, deposited materials penetrate deeper into deposited layers, resulting in a greater nozzle-to-surface gap, whereas lower gravity conditions reduce the penetration, allowing the layers to maintain their shape and remain closer to the nozzle, see e.g. printing on Phobos. The deformation occurs because the applied pressure exceeds the yield stress of the material. As mentioned earlier, two pressure contributions are involved: the hydrostatic pressure due to the weight of the deposited material and the extrusion pressure at the nozzle exit, which is applied to the printed layer when the material is deposited [30,49]. The hydrostatic pressure, P_H , applied to the bottom of the printed part is ρghN , where N is the number of deposited layers, ρ is the material density, and g is the gravitational acceleration of the celestial body. When considering two celestial bodies, such as the Moon and the Earth, the gravitational accelerations are 1.62 m/s^2 and 9.81 m/s^2 , respectively. The resulting hydrostatic pressures for a three-layered part are 121.2 Pa on the Moon and 751.2 Pa on Earth, while the material's yield stress is 41 Pa. Consequently, extreme deformation occurs when printed under Earth's gravity compared to printing on the Moon, where slight deformation is observed, cf. Fig. 4. Appendix Fig. 9 shows the comparison of cross-sectional shapes when each layer is printed and quantifies the percentile deformation of the bottom layer. It can be seen how previously printed layers deform when a new layer is added on top.

It can be concluded that the same material can be printed with geometrical precision and reduced deformation on low-gravity bodies but not on those with higher gravity. Different complexities, such as tuning the extrudability, may arise when printing in low-gravity environments using materials with higher rheological properties suitable for printing on Earth (high-gravity). Therefore, preparing compositions to have desired rheological properties and specifying printing conditions to control applied pressure require input based on the specific gravity of the printing environment.

3.2. 3D printing on the moon: influence of material's rheology and printing scale

When it comes to establishing lunar infrastructure using 3D printing, it is crucial to explore the possibility of achieving stable printing with no or controlled deformation under lunar gravity. This requires a detailed understanding of the influence of material rheology on the deformation of deposited layers as well as the potential to increase the printing scale. Two samples are considered to highlight the effect of material rheology while maintaining a constant printing scale of $D = 25$ mm. One sample contains a regolith weight fraction of 74.6%, and the other contains 77.3%, resulting in significant differences in yield stress (9.5 and 41 Pa, respectively) and slight variations in plastic viscosity and density, as detailed in Table 1. Notably, these yield stress values are relatively small, especially when compared to cementitious materials or geopolymers printed on Earth [33,35,63].

Fig. 5 presents the simulated multilayer deposition, the cross-sectional shapes at the middle of the deposited layers, and the comparison of cross-sectional shapes for successive layers. It can be seen that the deposited layers deform less as the regolith content increases due to the increase in yield stress. This trend, as expected, is the opposite of what is observed with increasing gravitational acceleration (presented earlier in Fig. 4), as the material's yield stress counteracts the applied pressure. In the presented cases, the hydrostatic pressures of the deposited parts differ slightly due to a small change in the densities of the mortars with the regolith weight fractions of 74.6% and 77.3%, as outlined in Table 1. The P_H values are 119.68 Pa and 124.05 Pa, whereas the yield stresses are 9.5 Pa and 41 Pa, respectively. Consequently, while the case with the highest regolith content retains its shape better, it is still expected to deform further as its yield stress remains lower than the hydrostatic pressure.

When fast printing capabilities are required, tailoring the printing scale is crucial. In this context, the most important processing parameters are the nozzle diameter and layer height, as they have the greatest influence on the applied pressure. Large-scale printing

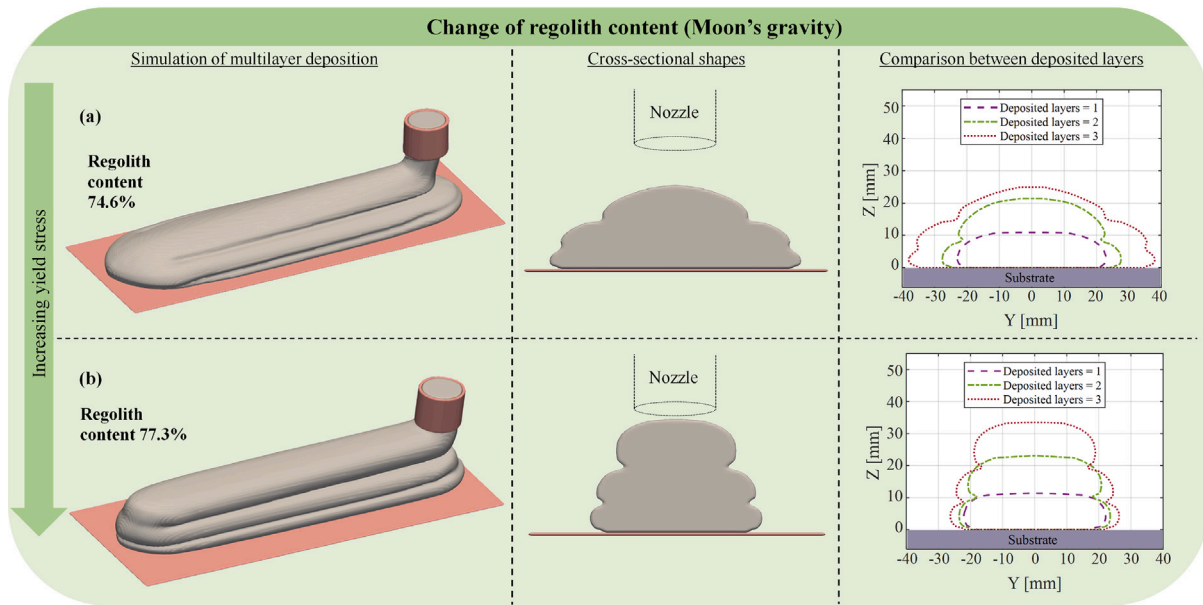


Fig. 5. Wet-on-wet printing of regolith on the Moon with different regolith weight fractions: (a) 74.6% and (b) 77.3%. The left column shows simulations of multilayer deposition, the middle column displays cross-sectional shapes at the midpoint of the deposited layers, and the right column illustrates changes in cross-sectional shapes for each successive layer.

significantly influences the hydrostatic pressure, whereas small-scale printing is dominated by the extrusion pressure [33].

A number of extrusion nozzles with $D = 5, 15, 25$, and 50 mm are considered to study the deformation and stability of the bottom layer at different printing scales, with the nominal layer height set to $D/2$. Furthermore, five different regolith mortars (cases 1–5) are considered, each with different regolith weight fractions (i.e., different yield stresses and plastic viscosities), see Table 1. For the individual effects of yield stress and plastic viscosity on layer deformation, as well as the influence of the printing and extrusion speeds, one can be referred to Refs. [30,33]. The percentile deformation of the deposited bottom layer is calculated when the second and third layers are printed for all the nozzle diameters and regolith weight fractions. The percentile deformation is presented in terms of interpolated color fields between nozzle diameters of 5 to 50 mm and yield stress between 9.5 to 41 Pa, see Fig. 6. Moreover, the cross-sectional shapes of deposited layers for all the investigated cases 1 to 5 in this section are presented in the

appendix, in Fig. 10, Fig. 11, Fig. 12 and Fig. 13 for different nozzle diameters.

Fig. 6-left shows the percentile of the deformation of the bottom layer when the second layer is printed. It can be observed that printing with the smallest investigated nozzle diameter of 5 mm results in smaller deformation for all investigated samples; however, deformation increases with the printing scale/nozzle diameter. Since the information in Fig. 6-left is constrained to instant deformation, it does not show the propagation of the deformation over time due to the hydrostatic pressure. To address this, a line is drawn at $P_H = \tau_0$. Above the line (when $P_H > \tau_0$), continuous deformation occurs, and the print cannot be stabilized without considering curing kinetics and the consequent yield stress buildup. Below the line (when $P_H < \tau_0$), deformation is caused instantly during extrusion by the total applied pressure, but it does not increase over time, as the hydrostatic pressure is smaller than the yield stress of the material. Therefore, the blue region below this line represents a stable zone where the printed layers will have very limited deformation during printing, which will remain unchanged

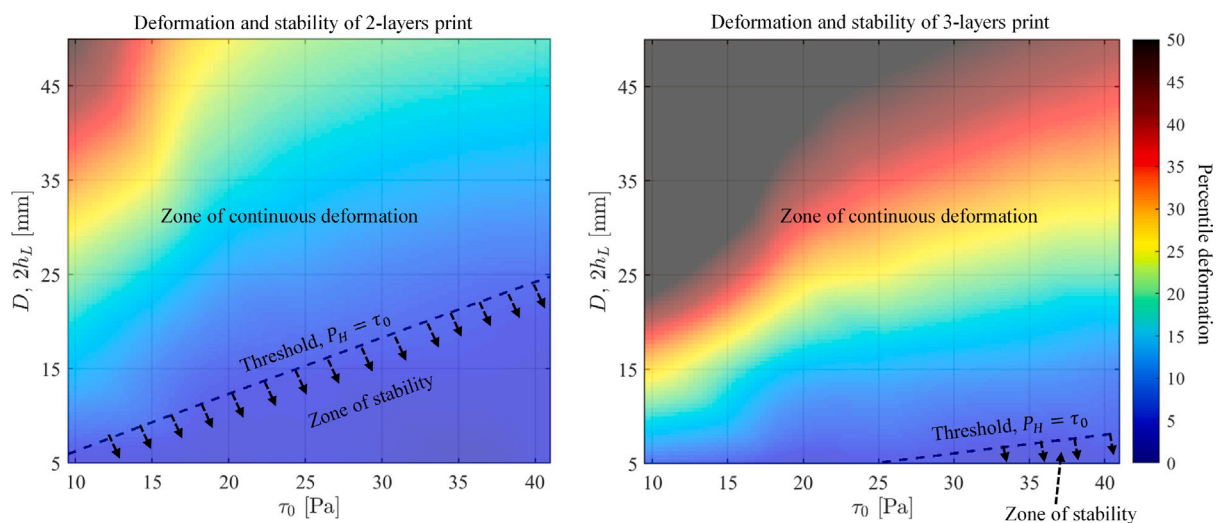


Fig. 6. Deformation and stability of multilayer printing as a function of nozzle diameter and yield stress of the regolith mortar for different regolith weight fractions.

over time. In the investigated cases, the stability zone offers a small window where the percentile deformation is approximately 10%, which supports printing with a nozzle diameter of 5 to 15 mm. A small stable zone is seen when printing with a nozzle diameter of 15 to 25 mm when the investigated yield stress is high; however, the stability zone cannot support the printing with a nozzle diameter larger than 25 mm due to the high hydrostatic pressure.

Fig. 6-right shows the bottom layer deformation when the third layer is printed. While the stability zone still exists for small nozzle diameters and higher regolith content, it has shrunk considerably. This is due to the newly added layer on top, which increased the hydrostatic pressure and made the zone of continuous deformation (where $P_H > \tau_0$) even larger. One could argue that increasing the yield stress of the material can stabilize the deformation; however, it leads to higher extrusion pressure and thereby cannot eliminate the deformation, as shown in [49]. This highlights the importance of considering and controlling the curing kinetics of regolith to minimize deformation and maximize printing scale, which can potentially enable the printing of more than three layers without structural collapse. Furthermore, as can be observed from the color bar, there is smaller deformation (below 10%) outside the stability zone. This suggests that if the curing kinetics could be adjusted to rapidly build-up the yield stress, thereby preventing continuous deformation, even an initially unstable print could be executed with reasonable geometric precision and accuracy. This insight prompted further analysis, taking into account the curing kinetics of the regolith, leading to the development of the wet-on-solidifying regolith model.

3.3. Curing of lunar regolith in fast and large scale printing

This section explores the wet-on-solidifying printing strategy, in which material deposition is closely coupled with its curing kinetics. Unlike the wet-on-semisolid printing strategy, this approach emphasizes not only the target yield stress at which the material should ultimately be cured, but also the rate at which yield stress builds up during curing. Such curing kinetics are well illustrated in geopolymer systems, where solidification is primarily governed by thermally activated reaction kinetics that are highly sensitive to curing temperature and composition, particularly the activator-to-binder ratio [64,65]. This approach allows the material to be deposited with a low yield stress, maintaining low extrusion pressure during deposition, and subsequently solidifying over time. Sufficient solidification can potentially be achieved during the travel of the nozzle and before printing the next layer on top. Consequently, the printing strategy is expected to support large-scale printing with good interlayer bonding between the

deposited layers. Furthermore, faster material curing can potentially mitigate evaporation issues observed in vacuum conditions [22].

The CFD model 2 has been utilized to study this crucial strategy, examining how deposited layers deform over time and determining the curing required to stabilize the printing process. The most critical behavior is expected to be observed during the deposition of two layers since the next layer is assumed to be added once the deposited first layer is stabilized. In this context, the model simulates the deposition of two layers using the time- and shear rate-dependent viscosity of the material. The model tracks material aging with a scalar approach and continuously increases the yield stress of the deposited material. The nozzle diameter is chosen to be $D = 50$ mm for large-scale printing, and the nominal layer height is $D/2$. The first layer is deposited with a length of $5D$. Further details about the CFD models and printing parameters are outlined in Table 2.

We simulate one minute of the curing after the deposition of each layer, where the model does not deposit new material, mimicking continuous deposition of fresh materials when printing large components as well as saving the computational cost. As the curing is modeled over time, the present framework does not account for lunar environmental effects, such as extreme temperature gradients or vacuum-induced evaporation and sublimation. These factors may significantly influence the curing kinetics. For example, Brandvold et al. [66] showed that the yield stress increases only gradually up to approximately 40 °C, but then rises nearly exponentially up to the study limit of 55 °C, suggesting that curing at daytime lunar temperatures could lead to significantly higher structural build-up rates than at room temperature. Furthermore, under vacuum conditions, evaporation and sublimation present additional challenges, as early work on sulfur concrete reported increased mass loss at elevated temperatures [67], and more recent investigations of geopolymers have documented severe water loss and even filament tearing during extrusion [23]. Therefore, these effects represent limitations of the current modeling framework and may be considered in future work.

Fig. 7 presents the bottom layer deformation over time. It is depicted that the nature of the deformation can be divided into three phases when a two-layer structure is printed and cured: curing of the first layer, deposition of the second layer, and curing after the deposition of the second layer. The deposition of the first layer is not included in one of these phases, as the initial measurement of the width takes place after its deposition. The initial printing yield stress is $\tau_0 = 41.35$ Pa, similar to that of the mortar with the largest regolith content studied in the previous section. The material cures, i.e., increases its yield stress $\tau_B(T_c)$, over the total printing time or curing time $T_c = 135.25$ s. Mathematically, $\tau_B(T_c) = \alpha T_c + \tau_0$, where α is the curing factor.

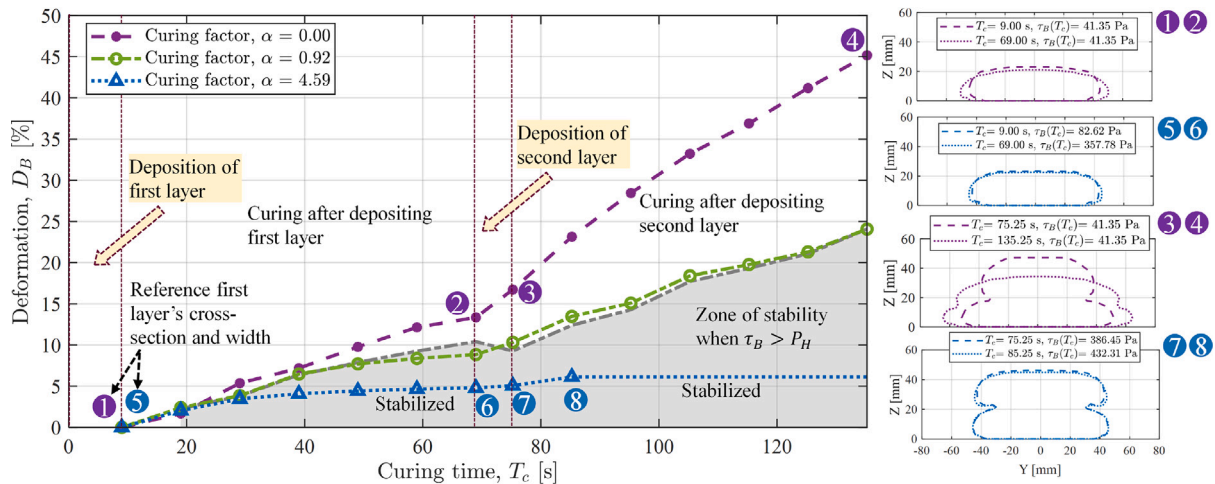


Fig. 7. Percentile Deformation of deposited bottom layer over curing time for different curing factors $\alpha = 0, 0.92$, and 4.59 Pa/s. Cross-sectional shapes are presented for wet-on-wet (no curing) and for wet-on-solidifying (highest investigated curing) printing in the Moon.

Three different curing cases are simulated: (i) no curing, i.e., wet-on-wet printing, where $\tau_B(T_c) = \tau_0$; (ii) low curing, where $\tau_B(T_c) = 1P_H$; and (iii) high curing, where $\tau_B(T_c) = 4P_H$. Here, $P_H = 165.4$ Pa represents the hydrostatic pressure of the two-layered part. Therefore, the corresponding curing factors can be calculated as $\alpha = (\tau_B(T_c) - \tau_0)/T_c$, and are 0, 0.92, and 4.59 Pa/s, respectively. These values fall within the range of curing factors reported in previous studies [68–70].

Fig. 7 shows that introducing curing to the material reduces the deformation of the bottom layer. In the absence of curing (i.e., wet-on-wet printing), the first layer's cross-sectional shape shows significant material spreading over time, resulting in a wider and thinner layer, and the second layer penetrates into the first layer and spreads further over time due to its low yield stress. When the curing factor is increased to 0.92 and 4.59 Pa/s, the material spreads more slowly than in the case with no curing due to the buildup of yield stress, cf. Appendix Fig. 14a, b. Particularly for the second layer, the shape is better maintained with fewer changes in width and height. Moreover, the changes in the cross-sectional shapes between the first layer right after deposition and the second layer after the waiting period highlight the improvements with different curing factors, as shown in Fig. 14c. It can be seen that the wet-on-solidifying strategy is able to retain the layered shape during large-scale printing with low-yield regolith, which is unachievable during wet-on-wet printing as seen in Fig. 6. Notably, with the highest investigated curing factor, the second layer forms a shape similar to the first, ensuring better surface roughness and stable behavior over time, cf. Fig. 7.

Although the exact measures of the bottom layer deformation depend on a complex interplay between the applied hydrostatic pressure, extrusion pressure, shape of the cross-section, and contact area between the layers, interesting observations can be made when compared with the hydrostatic pressure applied at the bottom of the layers. A gray zone is defined in Fig. 7 based on the gray curve of linearly interpolated deformation at $\tau_B = P_H$, using deformation values observed at τ_B near P_H . This gray zone indicates the stability region when $\tau_B > P_H$, while deformation occurs outside the gray zone when $\tau_B < P_H$.

It can be seen that the deformation increases over time with a higher slope when $\tau_B < P_H$ during the curing phases (after the deposition of both the first and second layers). Conversely, when $\tau_B > P_H$, the deformation stabilizes over time with a smaller slope (blue curve), where the observed deformation is below 5% within the gray zone. The case with a curing factor of 0.92 Pa/s (i.e., the green curve) moves in and out of the gray zone, initially remaining outside until about 45 s. Afterward, it enters the gray zone and stabilizes the first layer, but moves outside the gray zone again due to the addition of another layer on top, and therefore continues to deform over time. The last data point at the end of the second layer's curing phase meets the gray zone, and over time, the curve is expected to eventually flatten, similar to the blue curve, indicating the stabilization of the deformation. However, it takes a relatively long time to stabilize the deformation. Additionally, 25% deformation at the bottom layer is significant and will increase when another layer is added on top, as the hydrostatic pressure will also increase. Therefore, a curing factor of 0.92 Pa/s, where $\tau_B(T_c) = 1P_H$, is not sufficient for stable printing. Faster curing is required; for example, stable printing is observed in the investigated case with a curing factor of 4.59 Pa/s, where $\tau_B(T_c) = 4P_H$.

4. Conclusions

This study highlights the critical role of gravitational forces in the stability and geometrical precision of 3D-printed structures, as demonstrated through CFD modeling. We found that structures unstable under Earth's gravity may achieve stability or reduced deformation under lower-gravity extraterrestrial environments. Given the limited availability of experimental facilities in altered gravity conditions, theoretical tools like CFD modeling can be crucial for understanding off-Earth 3D printing.

The model simulates the wet-on-wet printing of lunar regolith mortar on the Moon for different regolith contents and printing scales (i.e., different nozzle diameters). It is observed that increasing the regolith content reduces the deformation of deposited layers as it increases the yield stress relative to the total applied pressures (the combined hydrostatic and extrusion pressures). In contrast, increasing the printing scale leads to significant deformation because the hydrostatic pressure increases with the height of the layers. The numerical results define a zone of stability, where the mortar's yield stress exceeds the hydrostatic pressure of the deposited part. Within the stability zone, the bottom layer experiences relatively low deformation, which is caused instantly during the extrusion. Outside the zone, the deformation becomes continuous as the hydrostatic pressure exceeds the material's yield stress. However, the stability zone diminishes as additional layers are deposited. Printing simply with high-yield stress regolith mortar is not an ideal solution for controlling deformation, as it can lead to higher extrusion pressures. Therefore, the printing process, particularly for large-scale printing, is required to utilize the curing kinetics of the mortar.

Our novel CFD model simulates wet-on-solidifying printing on the Moon and demonstrates that it is not necessary to print strictly within the zone of stability. Instead, low-yield material can be deposited, and the resulting deformation can be effectively controlled through the material's curing to achieve stable prints with reduced deformation. Results show that appropriate curing allows for large-scale printing with a nozzle diameter of 50 mm while still controlling the deformation of the bottom layer to below 5%. By utilizing this printing strategy (wet-on-solidifying), we can potentially enhance interlayer bonding and improve mechanical properties while ensuring fast printing, stability, and geometric precision of the part. Moreover, curing can address potential issues, such as minimizing evaporation during deposition under vacuum conditions.

In future work, it would be of interest to incorporate heat transfer into the CFD model to account for the extreme temperature gradients experienced on the Moon, as well as the effects of vacuum conditions, which can influence evaporation and sublimation during the printing process. Additionally, the influence of curing on a wider range of material compositions could be investigated to support stable printing.

CRedit authorship contribution statement

Md Tusher Mollah: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Berin Šeta:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Jon Spangenberg:** Writing – review & editing, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

See Table 5 and Figs. 8–14.

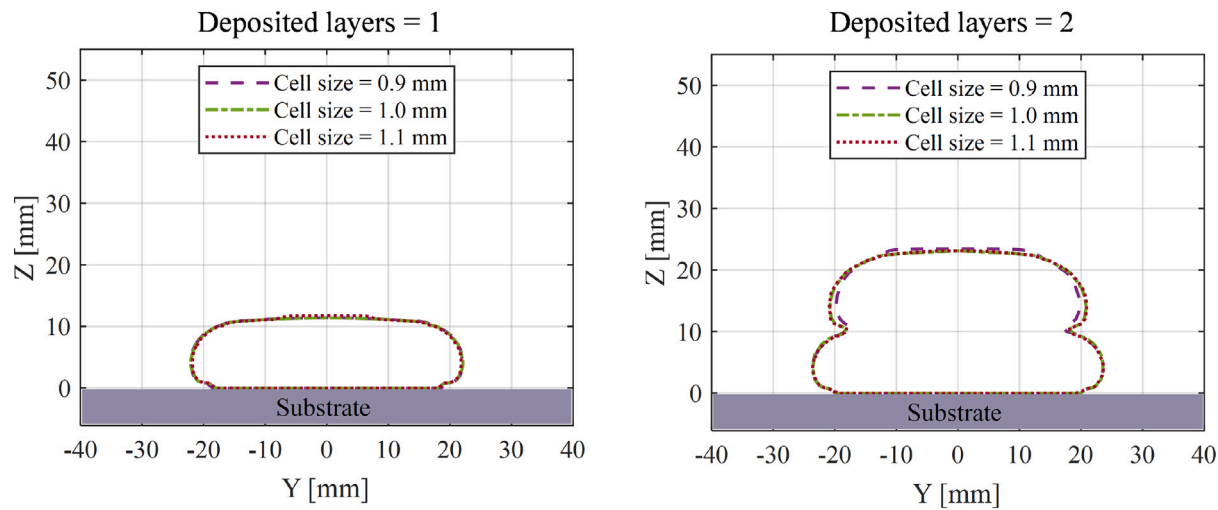


Fig. 8. Comparison of deposited layer cross-sectional shapes obtained using different mesh sizes.

Table 5
Layer widths and heights for different mesh resolutions.

Quantity	Layer	0.9 mm	1.0 mm	1.1 mm
Width (mm)	Layer 1	44.18	44.27	43.66
	Layer 2	47.14	47.19	47.36
Height (mm)	Layer 1	11.41	11.40	11.75
	Layer 2	23.41	22.93	22.99

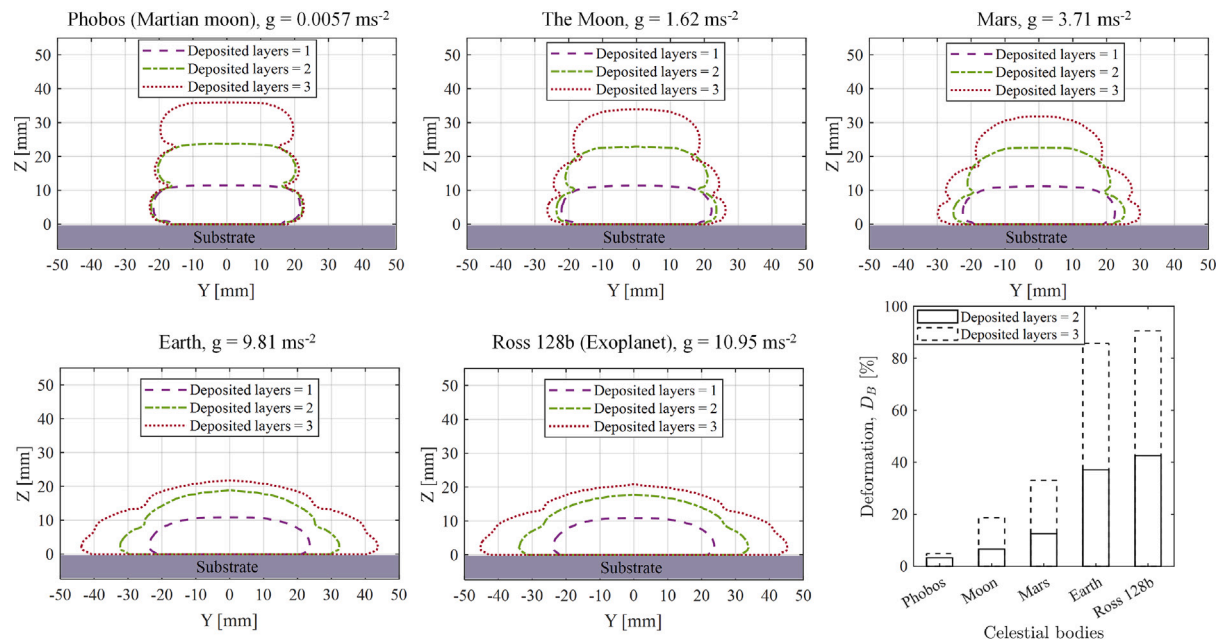


Fig. 9. Cross-sectional shapes of deposited layers as well as the percentile deformation of bottom layer when printed in different celestial bodies (change in gravity).

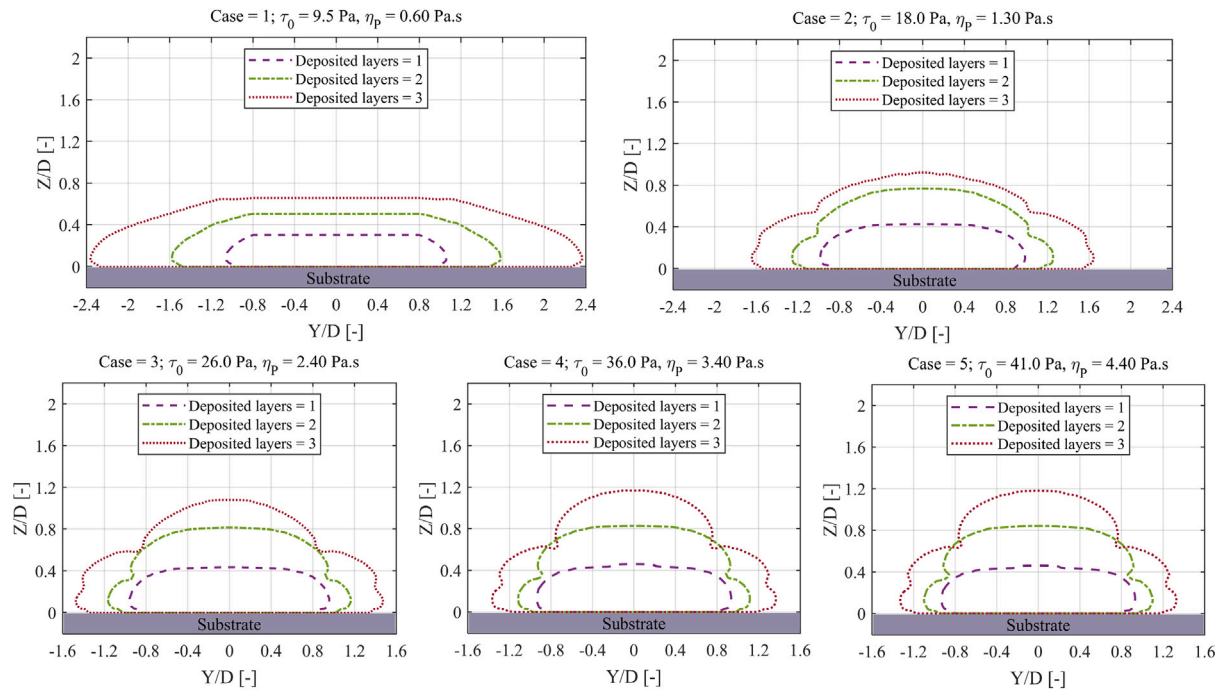


Fig. 10. Cross-sectional shapes of deposited layers with a nozzle diameter of 50 mm for different regolith content.

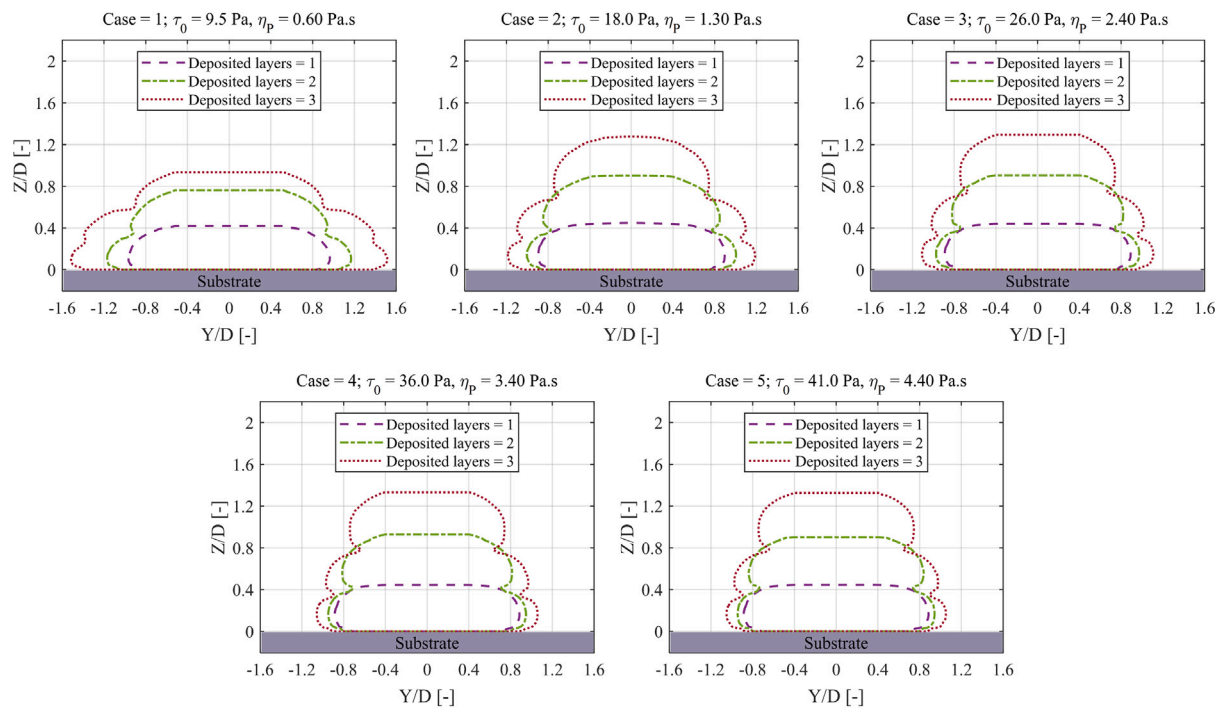


Fig. 11. Cross-sectional shapes of deposited layers with a nozzle diameter of 25 mm for different regolith content.

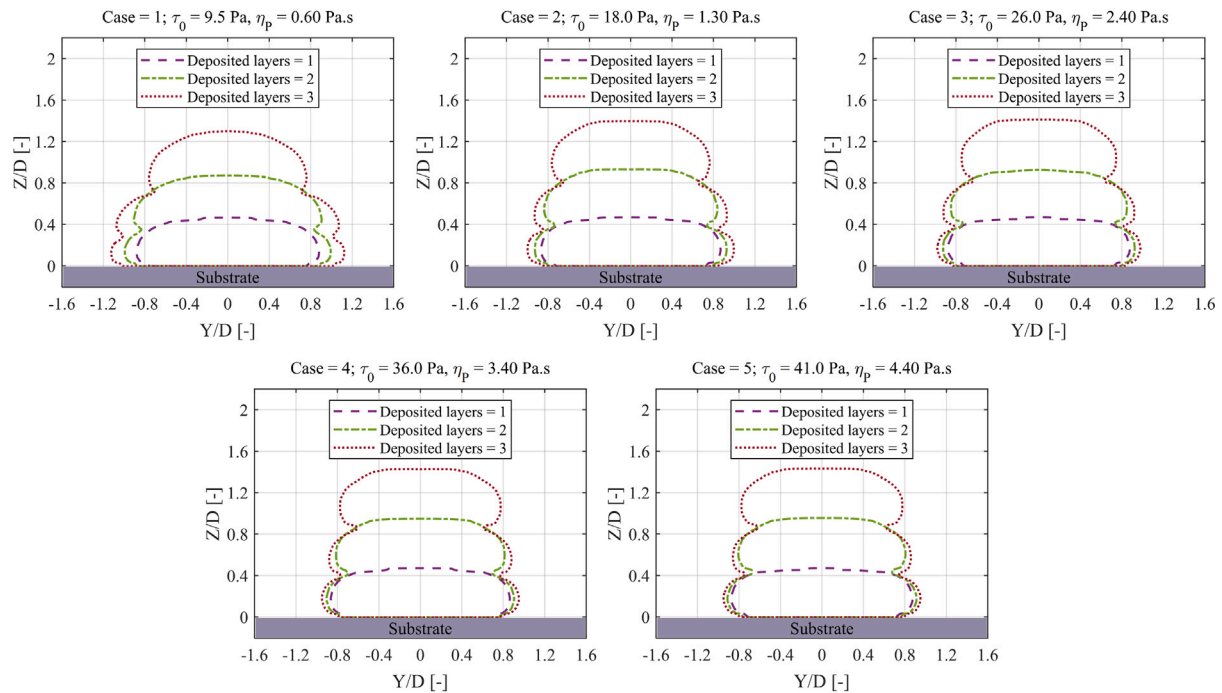


Fig. 12. Cross-sectional shapes of deposited layers with a nozzle diameter of 15 mm for different regolith content.

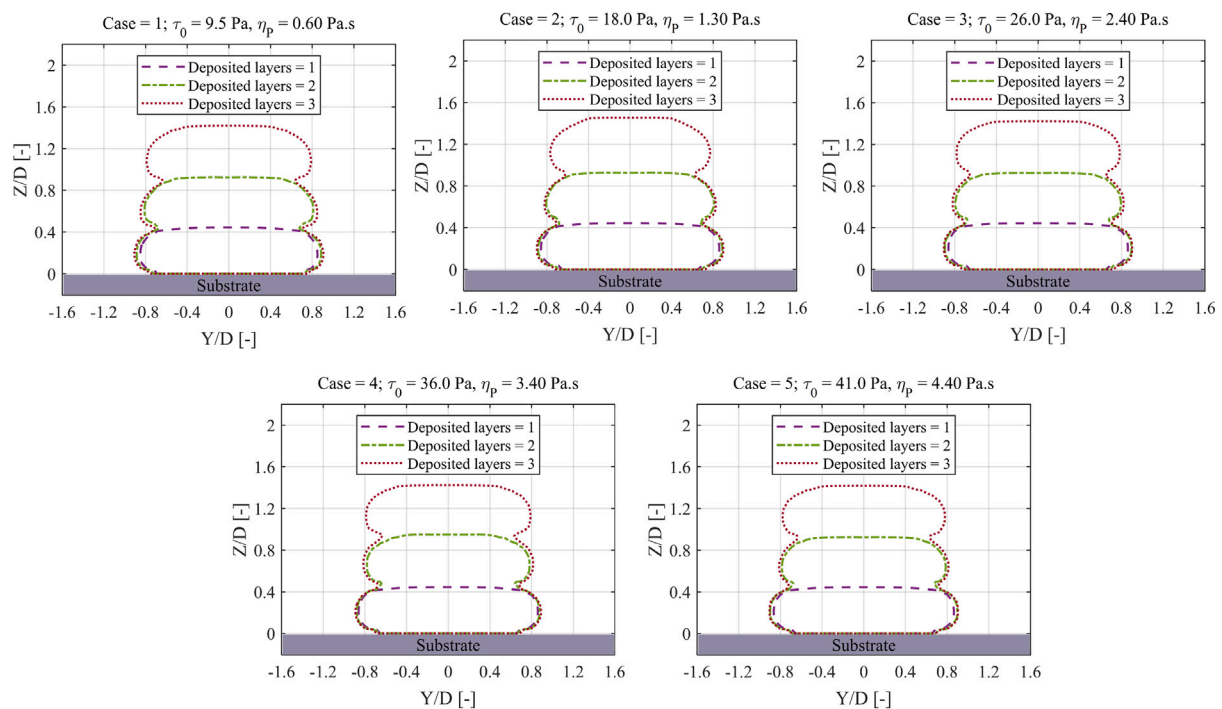


Fig. 13. Cross-sectional shapes of deposited layers with a nozzle diameter of 5 mm for different regolith content.

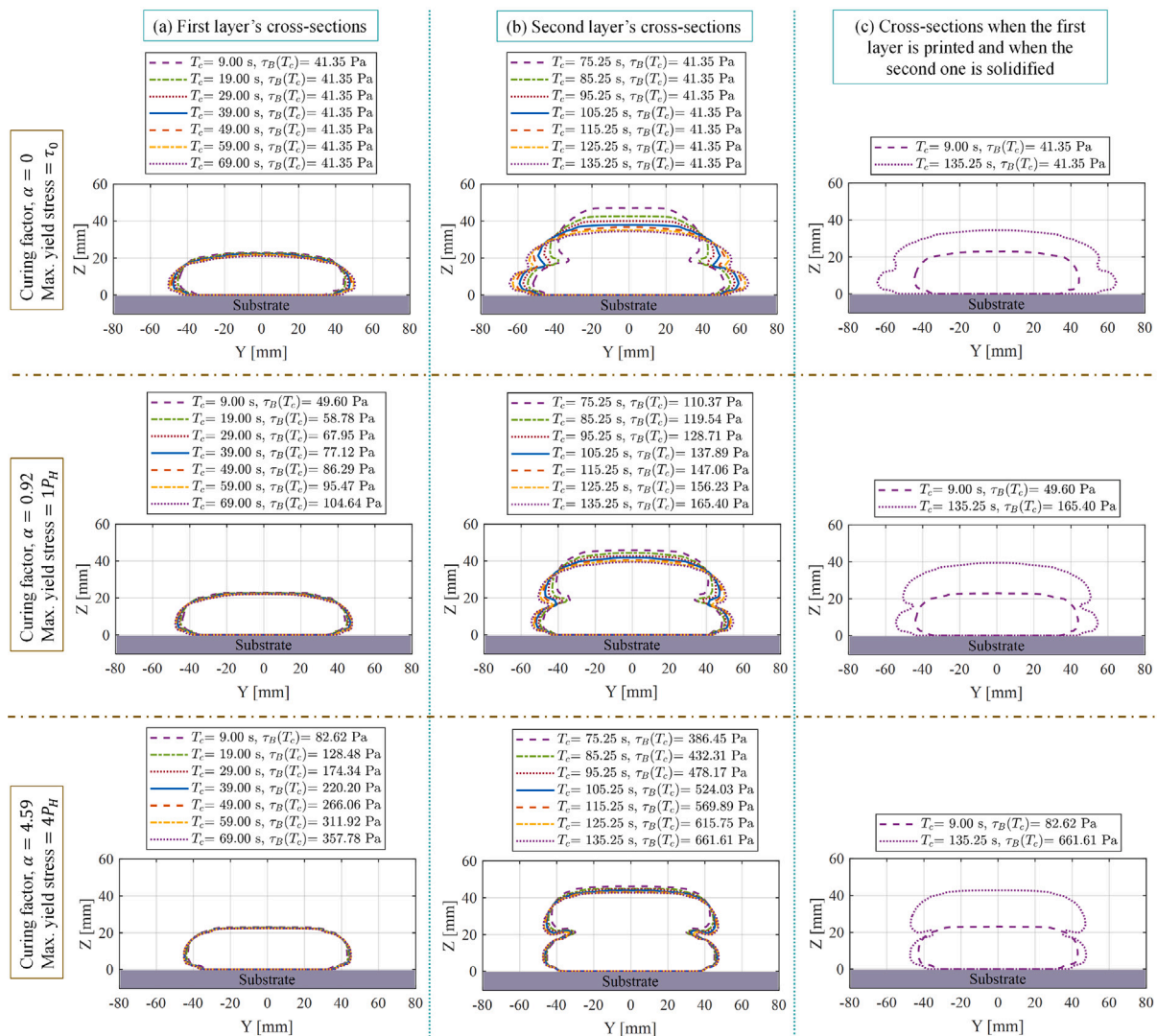


Fig. 14. Cross-sectional shapes of the deposited layers over curing time for different curing factors $\alpha = 0, 0.92$, and 4.59 result in maximum yield stresses equal to $\tau_0, 1P_H$, and $4P_H$, respectively. Cross-sectional shapes of: (a) the first layer for different building yield stresses; (b) the second layer for different building yield stresses; and (c) a comparison of the cross-section when the first layer is printed and when the second one gets solidified.

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